

Kalshi Prediction Market Model

Progress Report · Damien Nguyen · CSC 4999 · Advisor: Prof. Keith Mills

Overview

While rebuilding the pipeline for this semester, I found that the Spring dataset's price labels were mostly data errors, not real price movements. I corrected the underlying data, added new price-based features, and re-ran all evaluations. The corrected model now beats the momentum baseline; sentiment does not add a measurable improvement on top of price.

Data issues found and fixed

Issue	Effect on Spring dataset
Missing trade prices were recorded as \$0.00 instead of "no data"	70% of UP/DOWN labels were based on a \$0.00 price
The "4-hour" prediction window was measured in candles, not real time	Actual windows ranged from under 1 hour to several days (median 10 hours)
Momentum baseline compared events across different markets	Baseline was computed incorrectly for the first event in each market

All three were corrected in the new dataset. Prices now use the bid/ask midpoint, and each prediction window is exactly 4 real hours.

Results: before and after

Version	Model Macro F1	Momentum Baseline	Beats Momentum?
Spring, as published	0.418	0.395	Yes (result not reproducible — see note)
Spring, after leak fix	0.390	0.395	No
New dataset, price features only	0.499	0.455	Yes, statistically significant
New dataset, price + sentiment	0.507	0.455	Yes (sentiment adds no significant gain)

Note on the published result: the model and feature files saved from that run don't match each other, so the 0.418 score cannot be regenerated and is not used going forward.

New price-based features

Feature	Description
Mid price	Midpoint between best bid and ask
Spread	Best ask minus best bid

Feature	Description
Relative spread	Spread as a fraction of mid price
Distance from 0.5	How far the mid price is from an even-odds market
1h / 4h / 24h return	Price change over each lookback window
Previous outcome	The market's last resolved UP/DOWN/FLAT result
24h volatility	Standard deviation of hourly price changes
24h price range	High minus low over the past 24 hours
24h volume (log)	Trading volume over the past 24 hours
24h active hours	Number of hours with any trading activity
24h open-interest change	Change in outstanding contracts
Hours since market opened	Market age at prediction time

Sentiment feature results

News-sentiment coverage is currently limited to 2023–2024 markets (article data for 2025–2026 markets is pending). Within that coverage:

Comparison	Price Only	Price + Sentiment	Improvement
Events with news coverage	0.434	0.450	Not statistically significant
Sentiment features alone	—	0.338 (below momentum)	—

This is consistent with the Spring finding: news sentiment does not add measurable predictive value beyond price and market-structure information.

Dataset overview

	Count
Total markets	766
Total labeled events	21,587
Development (training/validation) events	18,335
Held-out test events (not yet evaluated)	1,965
Date range	Oct 2023 – Sep 2026

Outstanding item

News-article collection for 2025–2026 markets is paused pending renewed API access; all results above use available 2023–2024 sentiment data only.